Identifying Key Predictors of Work-From-Home Burnout:

A Machine Learning Feature Importance Analysis

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1. Introduction

With more people working from home, employee burnout has become a significant problem for companies worldwide. While machine learning can already predict burnout risk with high accuracy, the actual value for companies comes from understanding which specific factors drive burnout and at what thresholds intervention is warranted. Existing literature predicts burnout but does not explain which behaviors cause it or at what thresholds to intervene. This project directly address that gap.

This project investigates feature importance in work-from-home burnout prediction, identifying which behavioral factors (work hours, screen time, breaks, meetings, sleep, task completion, isolation, app switches) best predict risk and establishing evidence-based intervention thresholds. Rather than optimizing for prediction accuracy, this work optimizes for interpretability and actionability, restricting features to behavioral inputs only and explicitly excluding leakage prone variables.

Primary Research Question: Which work-from-home behaviors are the best predictors of burnout risk, and what intervention thresholds can be derived from the model outputs?

Secondary research questions address: (1) whether feature importance rankings are consistent across multiple classifiers; (2) whether a reduced-feature model can maintain comparable predictive performance; and (3) whether combinations of behavioral features predict burnout better than individual features alone. Findings from this analysis have direct applications in HR analytics, employee wellness program design, and employee burnout risk monitoring.

2. Literature Survey

According to Maslach and Leiter (2016), burnout is characterized by exhaustion, cynicism (depersonalization), and reduced personal accomplishment, with workload identified as the biggest contributing factor [1]. Burnout is commonly measured using the Maslach Burnout Inventory (MBI), which tracks burnout across these three characterizations. Alarcon's (2011) meta-analysis confirmed that job demands are strong predictors of burnout, but studies often use self-reported surveys rather than objective behavioral data, limiting their applicability to real-time monitoring [2].

In the context of remote work, Garfin et al. (2020) highlighted a correlation between increased screen time and elevated stress during COVID-19 remote work periods, suggesting that digital behaviors carry meaningful predictive power for occupational stress [3]. Sleep research further establishes that 7 to 8 hours of sleep per night is optimal for mental wellbeing, with deficits below this range associated with elevated psychological risk [4].

Machine learning applications for detecting occupational burnout are emerging but remain limited. Much of the existing work focuses on characterizing sample populations or maximizing prediction accuracy rather than explaining why burnout has occurred [5]. Shatte et al. (2019) demonstrated that behavioral data can predict mental health outcomes, though these studies also prioritize accuracy over interpretability [6]. Few studies translate model outputs into interpretable behavioral indicators or actionable intervention thresholds, particularly in remote work environments.

This project aims to shift the objective from simply predicting burnout to identifying feature importance patterns and deriving intervention thresholds from work-from-home behavioral data.

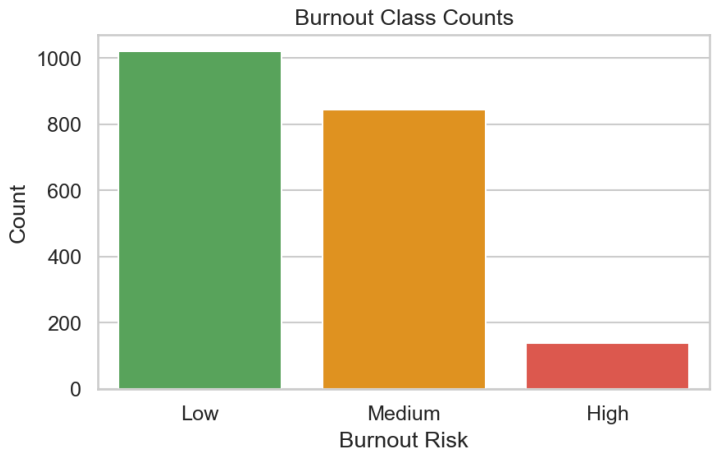
3. Methodology

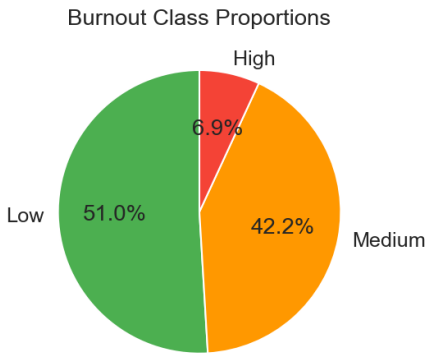
3.1 Dataset

The dataset contains 2,000 employee-day records from 150 remote workers tracked over 90 days. Records are described by behavioral features including work_hours, screen_time_hours, meetings_count, breaks_taken, after_hours_work, app_switches, sleep_hours, task_completion, isolation_index, and day_type (Weekday/Weekend). The data was auto-logged for the most part, but sleep_hours, isolation_index, and breaks_taken were all self-reported metrics. The target variable is burnout_risk with three classes: Low (51%), Medium (42%), and High (7%).

Two columns were identified as leakage variables and excluded from all models: fatigue_score and burnout_score. Both are derived constructs directly correlated with the burnout_risk target, so including them would produce artificially high accuracy while providing no causal insight. The final feature set comprises 9 behavioral features plus a binary encoding of day_type (Weekday=1, Weekend=0).

Class imbalance was addressed by using class_weight='balanced' in all three classifiers, which adjusts the loss function to penalize minority class errors more heavily. The High burnout class, being the smallest and most clinically important group, was prioritized in evaluation.





3.2 Data Preprocessing

Preprocessing was consistent across all three models. The day_type column was encoded as a binary integer (Weekday=1, Weekend=0). A stratified 80/20 train-test split with random_state=42 preserved class proportions across both sets, which was critical given the small minority High risk class. StandardScaler was fitted exclusively on training data and applied to the test set, preventing any test set statistical properties from influencing feature scaling.

3.3 Models and Feature Importance Methods

Three classifiers were trained, each using a different approach to measuring feature importance:

- Logistic Regression (LR): A linear model (max_iter=1000, class_weight='balanced'). Feature importance is computed as the mean absolute standardized coefficient across output classes, capturing direction and linear magnitude of each feature's contribution.
- Random Forest (RF): An ensemble of 200 decision trees (n_estimators=200, class_weight='balanced'). Feature importance is extracted via Gini impurity reduction (feature_importances_), capturing non-linear and tree-based relationships.

- Support Vector Machine (SVM): RBF-kernel SVM (class_weight='balanced'). Permutation importance was computed on the test set. Each feature was shuffled and the drop in ROC-AUC recorded, providing a model-agnostic validation of LR and RF rankings.

3.4 Threshold Analysis

For each of the top consensus features, intervention thresholds were derived as the midpoint between the Low burnout and High burnout class means.

3.5 Bootstrap Robustness Validation

To assess whether feature importance rankings are stable beyond a single train-test split, 100-iteration bootstrap resampling was performed. In each iteration, a dataset of size n=2,000 was drawn with replacement, a Random Forest was trained, and feature importances were recorded.

3.6 Simplified Model and Interaction Effects

Simplified models using the top 3, 4, and 5 consensus features were evaluated against the full 10-feature baseline using 5-fold cross-validated ROC-AUC. Three engineered interaction features were also tested: overwork_fatigue (work_hours × (10 − sleep_hours)), isolated_overwork (isolation_index × work_hours), and no_recovery (work_hours ÷ (breaks_taken + 1)).

4. Implementation

All analysis was implemented in Python using four sequential Jupyter notebooks. Core libraries include: pandas and numpy for data manipulation; scikit-learn for model training, evaluation, and preprocessing; matplotlib and seaborn for visualization; scipy.stats for Mann-Whitney U tests; and tqdm for progress tracking during bootstrap iterations.

A master main.py script executes key aspects of all four notebooks.

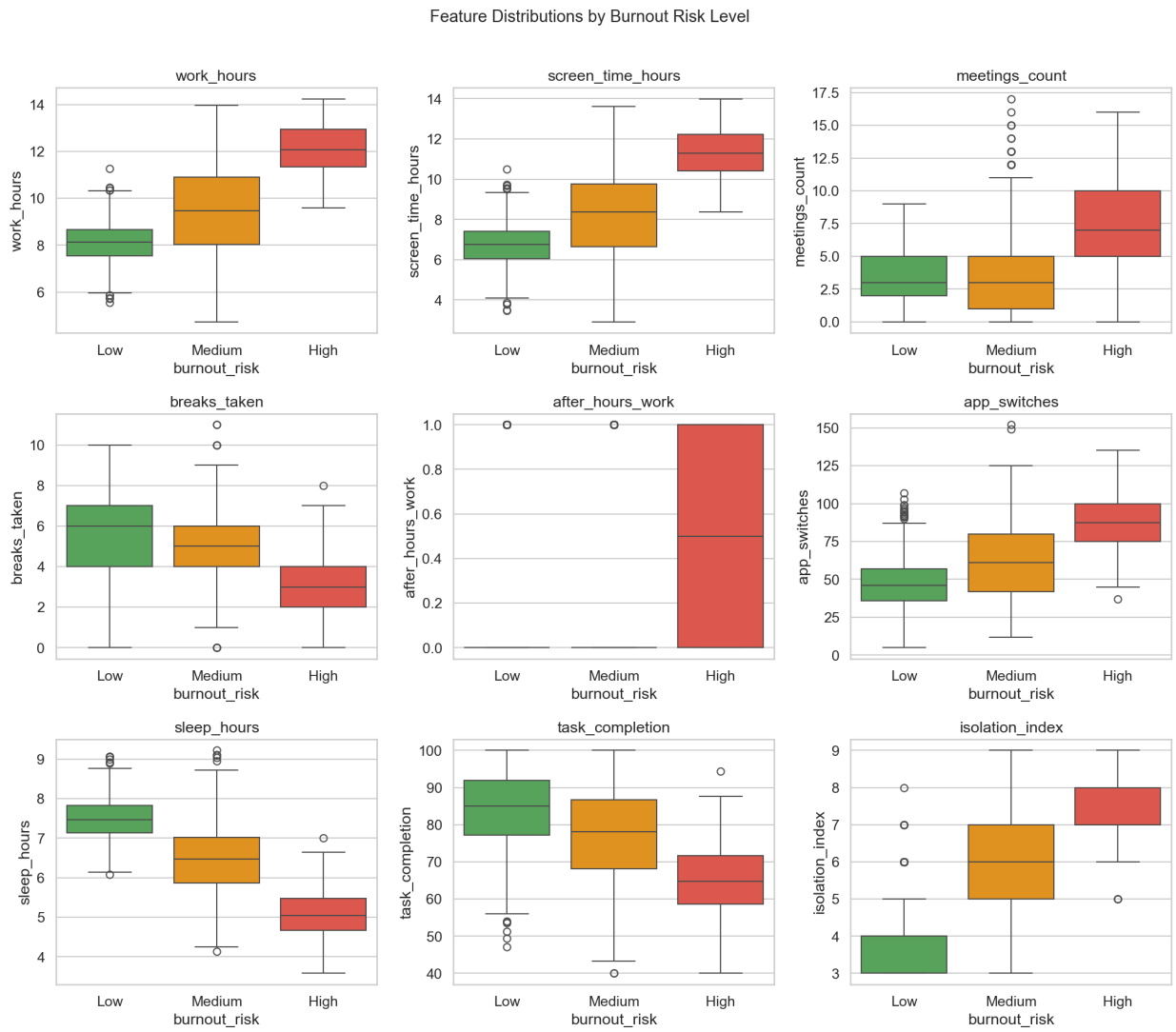
5. Experimental Results

5.1 Exploratory Data Analysis

The dataset contained no missing values and no duplicate rows. The class distribution showed a moderately imbalanced split across the three burnout risk levels. Low burnout accounted for 51% of records, Medium 42% and High 7%, yielding 138 High burnout cases.

Correlation analysis revealed the following top features by absolute correlation with the burnout_risk target (encoded as 0/1/2): isolation_index ($|r| = 0.71$), sleep_hours ($|r| = 0.70$), work_hours ($|r| = 0.60$), and screen_time_hours ($|r| = 0.59$).

Box plots confirmed visual separation across burnout classes for the highest-ranked features, with High-burnout individuals showing markedly different distributions. Kruskal-Wallis tests confirmed statistically significant differences across all three classes for the top features: isolation_index ($H = 1046.9$, $p < 0.001$), sleep_hours ($H = 910.2$, $p < 0.001$), work_hours ($H = 573.5$, $p < 0.001$), and screen_time_hours ($H = 575.1$, $p < 0.001$).



5.2 Model Performance

Table 1 summarizes classification performance for all three models on the 20% test set. A key metric of interest is recall on the High burnout class, as missing a high-risk individual (false negative) is more costly than a false alarm.

Table 1. Classification Performance Summary

Model	Accuracy	High Precision	High Recall	High F1	ROC-AUC (OvR)
Logistic Regression	*0.91*	*0.75*	*1.00*	*0.86*	*0.9842*
Random Forest	*0.91*	*0.84*	*0.78*	*0.81*	*0.9833*
SVM (RBF)	*0.91*	*0.64*	*1.00*	*0.78*	*0.9886*

All three models achieved identical overall accuracy (0.91), but diverge on the High burnout class, which is the most critical for intervention purposes. Logistic Regression and SVM both achieved perfect High burnout class recall

(1.00), meaning neither model missed a single at-risk employee in the test set. However, LR does so with higher precision (0.75 vs. 0.64), producing fewer false alarms. For a burnout screening tool, LR is therefore the preferred model. In a wellness context, failing to flag a burnt out employee is a far more costly error than an unnecessary check in, and LR achieves that safety net with greater efficiency than SVM. Random Forest achieved the highest precision on the High class (0.84) but at the cost of missing 22% of actual High burnout cases (recall 0.78), making it better suited for confirmatory analysis rather than initial screening.

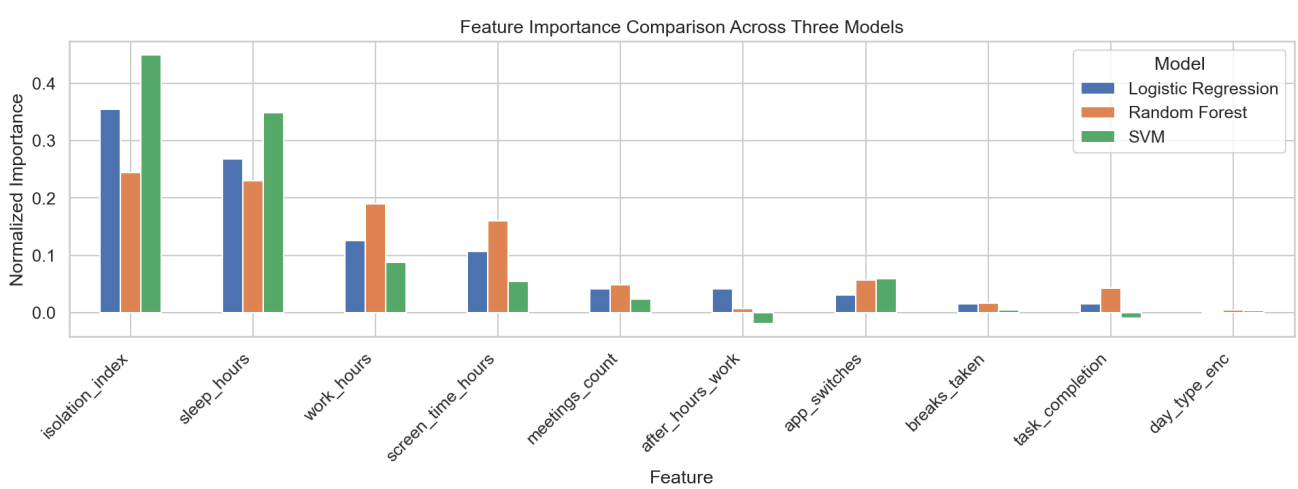
5.3 Feature Importance Comparison Across Models

Table 2 presents the feature importance rankings from all three models alongside the consensus mean rank. Comparing rankings across all three classifiers, which use fundamentally different mathematical approaches, identifies which features are robustly important regardless of the modeling method chosen.

Table 2. Feature Importance Rankings Across All Three Models

Feature	LR Rank (|coef|)	RF Rank (Gini)	SVM Rank (Perm.)	Mean Rank	Consensus
isolation_index	*1*	*1*	*1*	*1.0*	*#1*
sleep_hours	*2*	*2*	*2*	*2.0*	*#2*
work_hours	*3*	*3*	*3*	*3.0*	*#3*
screen_time_hours	*4*	*4*	*5*	*4.3*	*#4*
app_switches	*7*	*5*	*4*	*5.3*	*#5*
meetings_count	*5*	*6*	*6*	*5.7*	*#6*
breaks_taken	*8*	*8*	*7*	*7.7*	*#7*
task_completion	*9*	*7*	*9*	*8.3*	*#8 (tie)*
after_hours_work	*6*	*9*	*10*	*8.3*	*#8 (tie)*
day_type_enc	*10*	*10*	*8*	*9.3*	*#10*

All three models agreed unanimously on the top three predictors: isolation_index (#1), sleep_hours (#2), and work_hours (#3), each ranked identically across Logistic Regression, Random Forest, and SVM. The most significant finding is the dominance of isolation_index, which all three models identified as the single strongest behavioral predictor of burnout risk, challenging prior literature (Alarcon, 2011) that emphasized workload as the primary driver. Notably, day_type_enc (weekday vs. weekend) ranked last or near last across all three models, suggesting that when one works matters far less than how much and under what social and behavioral conditions work occurs.



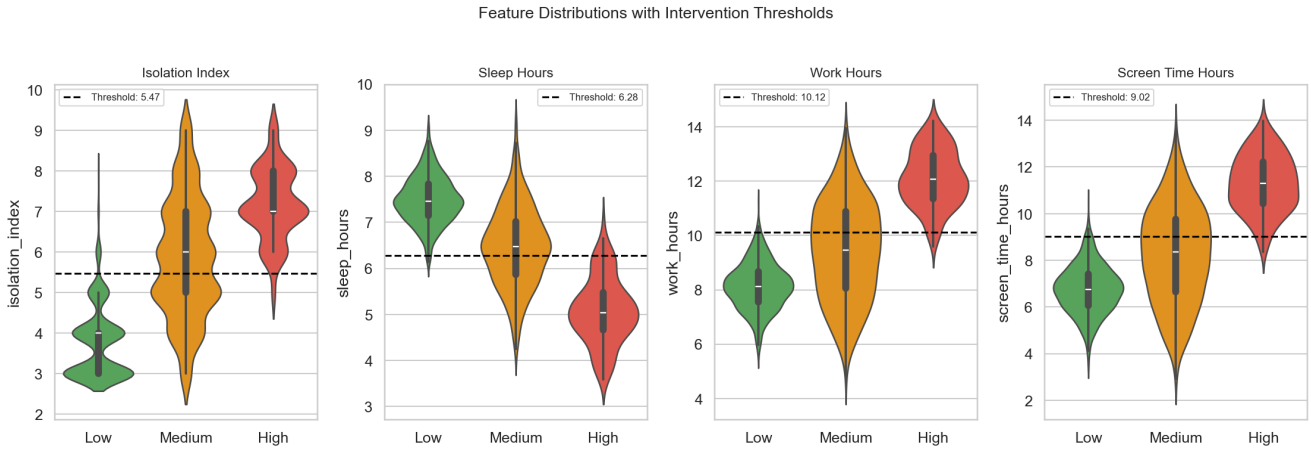
5.4 Intervention Thresholds

Table 3 presents statistically derived intervention thresholds for the top consensus features, validated by pairwise Mann-Whitney U tests. Thresholds are derived as the midpoint between the Low-burnout and High-burnout class means.

Table 3. Derived Intervention Thresholds

Feature	Low Mean	High Mean	Threshold	p-value	HR Intervention Flag
isolation_index	*3.8*	*7.2*	*5.5*	*p < 0.001*	*Flag if > 5.5*
sleep_hours	*7.5*	*5.1*	*6.3*	*p < 0.001*	*Flag if < 6.3*
work_hours	*8.1*	*12.1*	*10.1*	*p < 0.001*	*Flag if > 10.1*
screen_time_hours	*6.8*	*11.3*	*9.0*	*p < 0.001*	*Flag if > 9.0*

The derived thresholds provide concrete, actionable criteria for HR monitoring systems that go beyond simply predicting burnout risk. Most notably, the isolation_index threshold of 5.5 identifies social disconnection — not workload — as the primary intervention trigger, suggesting that employees reporting moderate-to-high isolation should be prioritized for wellness outreach before burnout escalates. The sleep_hours threshold of 6.3 hours per night aligns closely with clinical sleep medicine guidelines identifying fewer than 7 hours as a health risk (Watson & Badr, 2015), lending external validity to the derived threshold. Together, the four thresholds form a simple, deployable screening rule: flag any employee exceeding 10.1 daily work hours, 9.0 hours of screen time, an isolation score above 5.5, or sleeping fewer than 6.3 hours per night for proactive intervention.



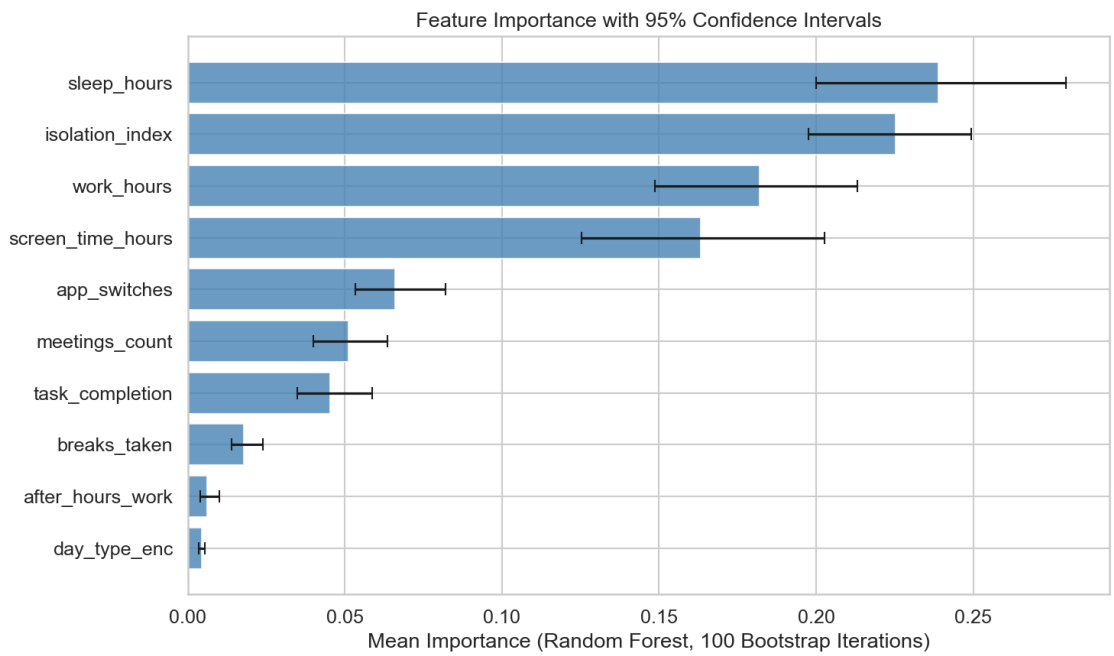
5.5 Bootstrap Robustness

To confirm that feature importance rankings are not artifacts of a single train-test split, 100-iteration bootstrap resampling was performed. Table 4 presents the mean importance and 95% confidence interval for each top feature across all iterations.

Table 4. Bootstrap Robustness — Feature Importance Stability (100 Iterations)

Feature	Mean Importance	95% CI Lower	95% CI Upper
#1 sleep_hours	*0.2389*	*0.1999*	*0.2795*
#2 isolation_index	*0.2252*	*0.1975*	*0.2495*
#3 work_hours	*0.1821*	*0.1486*	*0.2132*
#4 screen_time_hours	*0.1632*	*0.1254*	*0.2026*

Feature importance rankings remained stable across all 100 bootstrap iterations, with isolation_index and sleep_hours consistently occupying the top two positions and work_hours and screen_time_hours holding third and fourth. While adjacent confidence intervals show some overlap, particularly between sleep_hours (0.1999–0.2795) and isolation_index (0.1975–0.2495), and between work_hours and screen_time_hours, the top four features are clearly separated from the remaining six, confirming that the consensus ranking is not an artifact of any single train-test split.



5.6 Simplified Model Performance

Table 5 compares predictive performance of simplified models using only the top-ranked features against the full 10-feature baseline, evaluated via 5-fold cross-validated ROC-AUC.

Table 5. Simplified Model Performance

Model	# Features	CV ROC-AUC (Mean ± Std)	Test ROC-AUC
Full model (baseline)	10	*0.9842 ± 0.0035*	*0.9832*
Top 5 features	5	*0.9860 ± 0.0029*	*0.9825*
Top 4 features	4	*0.9845 ± 0.0032*	*0.9811*

Model	# Features	CV ROC-AUC (Mean ± Std)	Test ROC-AUC
Top 3 features	3	***0.9855 ± 0.0023***	***0.9837***
With interaction features	Top 4 + 3	***0.9873 ± 0.0028***	—

Reducing from 10 features to just 3 (isolation_index, sleep_hours, work_hours) resulted in no meaningful loss in predictive performance, the Top 3 model achieved a test ROC-AUC of 0.9837, marginally exceeding the full baseline (0.9832), demonstrating that 70% of input features are effectively redundant. The engineered interaction features (overwork_fatigue, isolated_overwork, no_recovery) showed high individual correlations with burnout_risk (r = 0.821, 0.753, and 0.431 respectively) but added only +0.0028 CV AUC over the top 4 baseline, a difference that shows no meaningful improvement, indicating that the individual features already capture the compounding effects sufficiently. For real world deployment, the Top 3 model is the recommended choice as it requires tracking only three behavioral variables, retains near-perfect performance, and is much simpler to implement and explain to HR stakeholders.

6. Conclusion and Future Research Directions

6.1 Summary of Findings

This project applied three machine learning classifiers to identify the key behavioral predictors of work-from-home burnout and derive actionable intervention thresholds. The analysis produced five clear findings:

First, isolation_index was the single most important predictor of WFH burnout, ranked #1 unanimously across Logistic Regression, Random Forest, and SVM. This challenges prior literature (Alarcon, 2011) which emphasized workload as the primary burnout driver.

Second, sleep_hours was the second most important predictor across all three models, with High-burnout employees averaging only 5.1 hours per night compared to 7.5 hours for Low-burnout employees. A threshold of 6.3 hours was found, which aligns with clinical sleep guidelines (Watson & Badr, 2015).

Third, the Top 3 feature model (isolation_index, sleep_hours, work_hours) achieved a test ROC-AUC of 0.9837, marginally exceeding the full 10-feature baseline of 0.9832, demonstrating that 70% of input features are redundant.

Fourth, engineered interaction features added no meaningful predictive value, indicating that individual behavioral features are sufficient.

Fifth, bootstrap resampling across 100 iterations confirmed that the top four feature rankings are stable and not artifacts of any single train-test split, with all four features clearly separated in importance from the remaining six.

6.2 Limitations

Several limitations warrant acknowledgment. The dataset contains a few self reported features, one of which is isolation_index, so the accuracy is dependent on employees perception of their own isolation. Causal direction for the top predictors cannot be established from cross-sectional data alone, for instance, social isolation may cause burnout, but burnout may also cause isolating behavior. Finally, the class imbalance in the High burnout group means threshold reliability would benefit from a larger sample of high burnout cases.

6.3 Future Research Directions

- Integrate objective behavioral metrics (e.g., calendar meeting frequency, communication platform activity, response latency) to replace or supplement self-reported scores.
- Build a real-time HR dashboard implementing the intervention threshold table, enabling proactive flagging of at-risk employees.
- Expand to longitudinal trajectory analysis to track individual employees over time and identify behavioral change patterns preceding burnout escalation or recovery.

References

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